

STRICHARTZ ESTIMATES FOR THE SCHRÖDINGER EQUATION

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1. INTRODUCTION

We will consider the linear homogeneous Schrödinger equation on $\mathbb{R}^n \times [0, \infty)$ given by $\partial_t u = i\Delta u$, subject to the initial condition $u(x, 0) = u_0(x)$. A solution is usually denoted by $u(x, t) = e^{it\Delta} u_0(x)$. If $2 \leq q, r \leq \infty$, $2/q + n/r = n/2$, and $(q, r, n) \neq (2, \infty, 2)$, then

$$\|e^{it\Delta} u_0\|_{L_t^q L_x^r} \lesssim \|u_0\|_{L_x^2(\mathbb{R}^n)},$$

where

$$\|u\|_{L_t^q L_x^r} = \left(\int_{\mathbb{R}} \|u(\cdot, t)\|_{L_x^r(\mathbb{R}^n)}^q dt \right)^{1/q}, \quad \|u\|_{L_{x,t}^q} = \|u\|_{L_t^q L_x^q}$$

and $f \lesssim g$ denotes the existence of some constant C (here dependent on q, r, n) such that $|f| \leq C|g|$. This is the general form of homogeneous Strichartz estimates for the linear Schrödinger equation. Strichartz originally showed such an estimate holds in the case $q = r = 2(n+2)/n$ for the non-homogeneous problem $\partial_t u = i(\Delta u + g)$ (see [4, Corollary 1]):

Theorem 1.1 (Strichartz, 1977). *If $u = e^{it\Delta} u_0$ solves $\partial_t u = i(\Delta u + g)$ with $u_0 \in L_x^2(\mathbb{R}^n)$ and $g \in L_{x,t}^p(\mathbb{R}^n \times \mathbb{R})$ where $p = 2(n+2)/(n+4) = q'$, then*

$$\|u\|_{L_{x,t}^q} \lesssim (\|u_0\|_{L_x^2(\mathbb{R}^n)} + \|g\|_{L_{x,t}^p}).$$

Here, we will focus our attention on this case with $g = 0$ using a proof presented in lectures by Larry Guth [2, Lectures 29–30]. For simplicity of the argument, we will also suppose $u_0 \in \mathcal{S}(\mathbb{R}^n)$, the Schwartz space.

Broadly speaking, these mixed Lebesgue norm estimates are of importance because of how they can provide a quantitative measure of the dispersion of solutions to various PDEs. Moreover, they have applications to problems of interest in the field, such as showing well-posedness for various equations (including nonlinear Schrödinger equations).

1.1. Fourier transform. The convention for the Fourier transform here is

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x} dx.$$

With this choice, Plancherel's theorem reads $\|\hat{f}\|_{L^2_\xi(\mathbb{R})} = \|f\|_{L^2_x(\mathbb{R})}$ and the normalization for Fourier inversion is

$$f(x) = (\hat{f})^\vee = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi.$$

Moreover, the Fourier transform interacts with differentiation as follows. Let $\alpha = (\alpha_1, \dots, \alpha_n)$ denote a multi-index, and define

$$x^\alpha = \prod_{i=1}^n x_i^{\alpha_i} \quad \text{and} \quad \partial^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

Then, under suitable conditions,

$$\partial^\alpha \hat{f} = [(-2\pi i x)^\alpha f]^\wedge \quad \text{and} \quad (\partial^\alpha f)^\vee(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi).$$

2. SOME PRELIMINARIES

We start by performing some formal computations without any conditions on the initial condition $u(x, 0) = u_0(x)$. Taking Fourier transforms on $\partial_t u = i\Delta u$, we obtain

$$\partial_t \hat{u}(\xi, t) = i(-4\pi|\xi|^2) \hat{u}(\xi, t).$$

Solving this ODE and using the initial condition $\hat{u}(\xi, 0) = \hat{u}_0(\xi)$, we get

$$\hat{u}(\xi, t) = e^{-4\pi i |\xi|^2 t} \hat{u}_0(\xi).$$

Then, after applying Fourier inversion we recover

$$u(x, t) = [e^{-4\pi i |\xi|^2 t} \hat{u}_0]^\vee(x) = e^{it\Delta} u_0(x).$$

From this process, we see that a desirable condition is $u_0 \in \mathcal{S}(\mathbb{R}^n)$, the Schwartz space, as then each step becomes valid, and we note that this implies our computed solution $u \in \mathcal{S}(\mathbb{R}^n)$. From now on, we always assume $u_0 \in \mathcal{S}(\mathbb{R}^n)$, so the derivation of the solution above applies.

2.1. Basic estimates. To obtain our desired Strichartz estimate, we will need some “endpoint estimates” followed by an estimate obtained by interpolation. We begin with the following conservation property.

Proposition 2.1. *The L^2 norm of a solution is preserved in t :*

$$\|e^{it\Delta} u_0\|_{L^2_x(\mathbb{R}^n)} = \|u_0\|_{L^2_x(\mathbb{R}^n)}.$$

Proof. This follows from applying Plancherel's theorem twice:

$$\begin{aligned} \|e^{it\Delta} u_0\|_{L^2_x(\mathbb{R}^n)} &= \|[e^{-4\pi i |\xi|^2 t} \hat{u}_0]^\vee\|_{L^2_x(\mathbb{R}^n)} = \|e^{-4\pi i |\xi|^2 t} \hat{u}_0\|_{L^2_\xi(\mathbb{R}^n)} \\ &= \|\hat{u}_0\|_{L^2_\xi(\mathbb{R}^n)} = \|u_0\|_{L^2_x(\mathbb{R}^n)}. \end{aligned}$$

□

Remark 2.2. One can show this holds for derivatives as well: $\|\partial_x^\alpha (e^{it\Delta} u_0)\|_{L^2_x(\mathbb{R}^n)}$ is constant in t .

Remark 2.3. There is no need for the initial data to be Schwartz here, and one can prove this without Fourier analysis. To do this, take real parts after integrating $(\partial_t u - i\Delta u)\bar{u} = 0$.

The following estimate gives decay for solutions to our Schrödinger equation.

Proposition 2.4. *We have the dispersion estimate*

$$\|e^{it\Delta}u_0\|_{L_x^\infty(\mathbb{R}^n)} \lesssim t^{-n/2}\|u_0\|_{L_x^1(\mathbb{R}^n)}.$$

Proof. By Fourier inversion and dominated convergence, we get

$$\begin{aligned} u(x, t) &= \int_{\mathbb{R}^n} \hat{u}(\xi, t) e^{2\pi i \xi \cdot x} d\xi = \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} e^{-\epsilon|\xi|^2} \hat{u}(\xi, t) e^{2\pi i \xi \cdot x} d\xi \\ &= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} e^{-\epsilon|\xi|^2} e^{-4\pi i |\xi|^2 t} \hat{u}_0(\xi) e^{2\pi i \xi \cdot x} d\xi \\ &= \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} \left(e^{-\epsilon|\xi|^2} e^{-4\pi i |\xi|^2 t} e^{2\pi i \xi \cdot x} \int_{\mathbb{R}^n} u_0(y) e^{-2\pi i \xi \cdot y} dy \right) d\xi. \end{aligned}$$

Using Fubini,

$$u(x, t) = \lim_{\epsilon \rightarrow 0} K_{t, \epsilon} * u_0 = \lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} K_{t, \epsilon}(x - y) u_0(y) dy$$

where

$$K_{t, \epsilon}(z) = \int_{\mathbb{R}^n} e^{-\epsilon|\xi|^2} e^{-4\pi i |\xi|^2 t} e^{2\pi i \xi \cdot z} d\xi,$$

so we have

$$\|e^{it\Delta}u_0(x)\|_{L_x^\infty(\mathbb{R}^n)} \leq \|u_0(x)\|_{L_x^1(\mathbb{R}^n)} \liminf_{\epsilon \rightarrow 0} \|K_{t, \epsilon}(z)\|_{L_z^\infty(\mathbb{R}^n)}.$$

To produce estimates of $K_{t, \epsilon}$, notice that

$$\int_{\mathbb{R}^n} e^{-\epsilon|\xi|^2} e^{-4\pi i |\xi|^2 t} e^{2\pi i \xi \cdot z} d\xi = \prod_{j=1}^n \int_{\mathbb{R}} e^{-\epsilon \xi_j^2} e^{-4\pi i \xi_j^2 t} e^{2\pi i \xi_j z_j} d\xi_j.$$

One checks (either by using the Gaussian formula for $e^{-a\xi^2}$ using a having positive real part, or splitting the integral appropriately to estimate) that

$$\left| \int_{\mathbb{R}} e^{-\epsilon \xi_j^2} e^{-4\pi i \xi_j^2 t} e^{2\pi i \xi_j z_j} d\xi_j \right| \lesssim t^{-1/2}$$

uniformly as $\epsilon \rightarrow 0$, so we are done. $\square$

Remark 2.5. One can explicitly compute the Schrödinger kernel

$$K_t(z) = \frac{1}{(4\pi i t)^{n/2}} e^{i \frac{|z|^2}{4t}}$$

which appeared in this process, and the solution to our PDE would then be given by $u = K_t * u_0$. This way, the dispersion estimate is easily obtained.

Remark 2.6. This estimate is sharp. If u_0 is a smooth bump supported on the unit ball B_1 in $\mathbb{R}^n$, then the above computation can be used to show that

$$\|e^{it\Delta}u_0\|_{L_x^\infty(\mathbb{R}^n)} \gtrsim t^{-n/2}\|u_0\|_{L_x^1(\mathbb{R}^n)}.$$

Using the preservation of the L^2 norm and this dispersion estimate, we can interpolate to obtain $L_x^p(\mathbb{R}^n) \rightarrow L_x^{p'}(\mathbb{R}^n)$ estimates in a wider range. First, we recall Riesz–Thorin interpolation theorem:

Theorem 2.7 (Riesz–Thorin interpolation). *Let X and Y be σ -finite measure spaces (under some implicit measure) and $1 \leq p_0, p_1, q_0, q_1 \leq \infty$. Suppose $T : L^{p_0}(X) + L^{p_1}(X) \rightarrow L^{q_0}(Y) + L^{q_1}(Y)$ is a linear operator that is bounded as an operator $L^{p_0}(X) \rightarrow L^{q_0}(Y)$ and $L^{p_1}(X) \rightarrow L^{q_1}(Y)$. Then, for all $\theta \in (0, 1)$, we have that T is bounded as a map $L^{p_\theta}(X) \rightarrow L^{q_\theta}(Y)$ where*

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1} \quad \text{and} \quad \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Moreover,

$$\|T\|_{L^{p_\theta}(X) \rightarrow L^{q_\theta}(Y)} \leq \|T\|_{L^{p_0}(X) \rightarrow L^{q_0}(Y)}^{1-\theta} \|T\|_{L^{p_1}(X) \rightarrow L^{q_1}(Y)}^\theta.$$

Applying Riesz–Thorin gives the following estimate.

Corollary 2.7.1. *Let $1/p = (1-\theta)/2$. Then,*

$$\|e^{it\Delta}u_0\|_{L_x^p(\mathbb{R}^n)} \lesssim t^{-n\theta/2}\|u_0\|_{L_x^{p'}(\mathbb{R}^n)}.$$

Proof. We have the endpoint estimates $\|e^{it\Delta}u_0\|_{L_x^2(\mathbb{R}^n)} \leq \|u_0\|_{L_x^2(\mathbb{R}^n)}$ for $p_0 = q_0 = 2$ and $\|e^{it\Delta}u_0\|_{L_x^\infty(\mathbb{R}^n)} \lesssim t^{-n/2}\|u_0\|_{L_x^1(\mathbb{R}^n)}$ for $p_1 = 1, q_1 = \infty$. For $\theta \in (0, 1)$, the Riesz–Thorin exponent p between p_1 and q_1 satisfies

$$\frac{1}{p} = \frac{1-\theta}{2} + \frac{\theta}{\infty} = \frac{1-\theta}{2},$$

and the $L_x^2(\mathbb{R}^n) \rightarrow L_x^2(\mathbb{R}^n)$ operator norm of $e^{it\Delta}$ is 1, so

$$\|e^{it\Delta}\|_{L_x^{p'}(\mathbb{R}^n) \rightarrow L_x^p(\mathbb{R}^n)} \leq \|e^{it\Delta}\|_{L_x^1(\mathbb{R}^n) \rightarrow L_x^\infty(\mathbb{R}^n)}^\theta \lesssim (t^{-n/2})^\theta = t^{-n\theta/2}.$$

This is precisely the operator bound we sought.

As a note of caution, we remark that the implicit constant here has dependence on θ . $\square$

2.2. Properties of $e^{it\Delta}$. We note two facts we will later use about the operator $e^{it\Delta}$ below.

Lemma 2.7.2. *$e^{it\Delta}$ defines a unitary operator on $L_x^2(\mathbb{R}^n)$.*

Proof. Compute

$$\begin{aligned}
\langle e^{it\Delta} f, g \rangle &= \int_{\mathbb{R}^n} [e^{it\Delta} f](x) \overline{g(x)} dx = \int_{\mathbb{R}^n} [e^{-4\pi i |\xi|^2 t} \hat{f}(\xi)]^\vee(x) \overline{g(x)} dx \\
&= \int_{\mathbb{R}^n} e^{-4\pi i |\xi|^2 t} \hat{f}(\xi) \overline{\hat{g}(\xi)} d\xi = \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{(e^{4\pi i |\xi|^2 t} \hat{g}(\xi))} d\xi \\
&= \int_{\mathbb{R}^n} f(x) \overline{[e^{4\pi i |\xi|^2 t} \hat{g}(\xi)]^\vee(x)} dx = \int_{\mathbb{R}^n} f(x) \overline{[e^{-it\Delta} g](x)} dx \\
&= \langle f, e^{-it\Delta} g \rangle.
\end{aligned}$$

using the unitarity of the Fourier transform on $L^2(\mathbb{R}^n)$ $\square$

Lemma 2.7.3. $e^{it\Delta}$ has the semigroup property

$$e^{is\Delta}(e^{it\Delta}u_0) = e^{i(s+t)\Delta}u_0.$$

Proof. We have

$$\begin{aligned}
e^{is\Delta}(e^{it\Delta}u_0) &= [e^{-4\pi i |\xi|^2 s} (e^{it\Delta}u_0)^\wedge]^\vee = [e^{-4\pi i |\xi|^2 s} (e^{-4\pi i |\xi|^2 t} u_0)^\wedge]^\vee \\
&= [e^{-4\pi i |\xi|^2 (s+t)} u_0]^\vee = e^{i(s+t)\Delta}u_0
\end{aligned}$$

by Fourier inversion. $\square$

3. PROOF OF STRICHARTZ ESTIMATES

We prove the Strichartz estimate

$$\|e^{it\Delta}u_0\|_{L_{x,t}^q} \lesssim \|u_0\|_{L_x^2(\mathbb{R}^n)} \quad (1)$$

with $q = 2(n+2)/n$ in this section. First, we start with a Duhamel principle formula for the inhomogeneous Schrödinger equation $\partial_t u = i\Delta u + f$. Then, we obtain a precursory Strichartz estimate using Hardy–Littlewood–Sobolev fractional integral theory, and afterwards are able to conclude the estimate (1) with the help of a duality trick.

Proposition 3.1 (Duhamel principle). *Suppose $f \in C_c^\infty(\mathbb{R}^n \times \mathbb{R})$. Then,*

$$u(x, t) = \int_{-\infty}^t e^{i(t-s)\Delta} f(x, s) ds$$

solves $\partial_t u = i\Delta u + f$. Moreover, let T be such that $(x, t) \notin \text{supp } f$ for all $t < T$. Then, for all $t < T$, $u(x, t) = 0$ (we then say that u vanishes before the support of f).

Proof. Let u be as defined above. Applying the Leibniz integral rule gives

$$\begin{aligned}
\partial_t u(x, t) &= e^{i(t-s)\Delta} f(x, s)|_{s=t} + \int_{-\infty}^x \partial_t (e^{i(t-s)\Delta} f(x, s)) ds \\
&= f(x, t) + \int_{-\infty}^x i\Delta (e^{i(t-s)\Delta} f(x, s)) ds \\
&= f(x, t) + i\Delta \left(\int_{-\infty}^x e^{i(t-s)\Delta} f(x, s) ds \right) \\
&= i\Delta u(x, t) + f(x, t).
\end{aligned}$$

Then we see that for $t < T$, $\partial_t u = 0$, and u is Schwartz so we conclude that $u(x, t) = 0$ for all $(x, t) \in \mathbb{R}^n \times (-\infty, T)$. $\square$

Using the Hardy–Littlewood–Sobolev fractional integral theorem stated below, we will obtain a precursory Strichartz estimate.

Theorem 3.2 (Hardy–Littlewood–Sobolev fractional integration). *Let α be such that $0 < \alpha < n$. If $1 < p < \frac{n}{\alpha}$ and $\frac{1}{q} = \frac{1}{p} - \frac{\alpha}{n}$, then the operator $f \mapsto K_\alpha * f$ where $K_\alpha(x) = |x|^{\alpha-n}$ is $L^p(\mathbb{R}^n) \rightarrow L^q(\mathbb{R}^n)$ bounded.*

Lemma 3.2.1 (Strichartz estimate, Duhamel solution). *Let $f \in C_c^\infty(\mathbb{R})$ and let u satisfy $\partial_t u = i\Delta u + f$. Also suppose that u vanishes before the support of f . Then,*

$$\|u\|_{L_{x,t}^q} \lesssim \|f\|_{L_{x,t}^{q'}}.$$

Proof. Minkowski gives

$$\|u(\cdot, t)\|_{L_x^q(\mathbb{R}^n)} \leq \int_{-\infty}^{\infty} \|e^{i(t-s)\Delta} f(x, s)\|_{L_x^q(\mathbb{R}^n)} ds.$$

Taking $\theta = 1 - \frac{2}{q}$, we can apply Corollary 2.7.1 and obtain

$$\|u(\cdot, t)\|_{L_x^q(\mathbb{R}^n)} \lesssim \int_{-\infty}^{\infty} (t-s)^{-n\theta/2} \|f(\cdot, s)\|_{L_x^{q'}(\mathbb{R}^n)} ds$$

which is a fractional Hardy–Littlewood–Sobolev fractional integral. One checks with our choice of θ , recalling $q = 2(n+2)/n$, that the conditions for the Hardy–Littlewood–Sobolev fractional integration theorem are met, and applying it gives the desired estimate. $\square$

With this, we now prove the Strichartz estimate (1):

Theorem 3.3. *If $u_0 \in \mathcal{S}(\mathbb{R}^n)$ and u solves the homogeneous Schrödinger problem*

$$\begin{cases} \partial_t u(x, t) = i\Delta u(x, t) & \text{for } t > 0, \\ u(x, 0) = u_0(x), \end{cases}$$

then

$$\|e^{it\Delta} u_0\|_{L_{x,t}^q} \lesssim \|u_0\|_{L_x^2(\mathbb{R}^n)} \quad (1)$$

where $q = 2(n+2)/n$.

Proof. Let $q' = 2(n+2)/(n+4)$ be the L^q dual exponent ($1/q + 1/q' = 1$), so duality gives

$$\|e^{it\Delta}u_0\|_{L^q_{x,t}} = \sup_{\|f\|_{L^{q'}_{x,t}}=1} \int_{\mathbb{R}} \int_{\mathbb{R}^n} [e^{it\Delta}u_0](x) \overline{f(x,t)} dx dt,$$

and since $e^{it\Delta}$ is unitary this is equal to

$$\sup_{\|f\|_{L^{q'}_{x,t}}=1} \int_{\mathbb{R}} \langle u_0, e^{-it\Delta}f(\cdot, t) \rangle dt = \sup_{\|f\|_{L^{q'}_{x,t}}=1} \langle u_0, \int_{\mathbb{R}} e^{-it\Delta}f(\cdot, t) dt \rangle.$$

Here, we may apply Cauchy–Schwarz to obtain

$$\|e^{it\Delta}u_0\|_{L^q_{x,t}} \leq \|u_0\|_{L^2_x(\mathbb{R}^n)} \sup_{\|f\|_{L^{q'}_{x,t}}=1} \left\| \int_{\mathbb{R}} e^{-it\Delta}f(\cdot, t) dt \right\|_{L^2_x(\mathbb{R}^n)}. \quad (2)$$

Now, observe that

$$\begin{aligned} \left\| \int_{\mathbb{R}} e^{-it\Delta}f(\cdot, t) dt \right\|_{L^2_x(\mathbb{R}^n)}^2 &= \left\langle \int_{\mathbb{R}} e^{-it\Delta}f(\cdot, t) dt, \int_{\mathbb{R}} e^{-is\Delta}f(\cdot, s) ds \right\rangle \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \langle e^{-it\Delta}f(\cdot, t), e^{-is\Delta}f(\cdot, s) \rangle dt ds \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \langle f(\cdot, t), e^{i(t-s)\Delta}f(\cdot, s) \rangle dt ds \\ &= \int_{\mathbb{R}} \langle f(\cdot, t), \int_{\mathbb{R}} e^{i(t-s)\Delta}f(\cdot, s) ds \rangle dt \end{aligned}$$

using the semigroup property. Next, writing

$$F(x, t) = \int_{\mathbb{R}} e^{i(t-s)\Delta}f(x, s) ds,$$

we have

$$\begin{aligned} \left\| \int_{\mathbb{R}} e^{-it\Delta}f(\cdot, t) dt \right\|_{L^2_x(\mathbb{R}^n)}^2 &= \int_{\mathbb{R}} \langle f(\cdot, t), F(\cdot, t) \rangle dt \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}^n} f(x, t) \overline{F(x, t)} dx dt. \end{aligned}$$

Using Hölder's inequality for $L_{x,t}$ norms, we have

$$\int_{\mathbb{R}} \int_{\mathbb{R}^n} f(x, t) \overline{F(x, t)} dx dt \leq \|f\|_{L^{q'}_{x,t}} \|F\|_{L^q_{x,t}}.$$

But F is a solution for which the Duhamel solution estimate Lemma 3.2.1 applies to get $\|F\|_{L_{x,t}^q} \lesssim \|f\|_{L_{x,t}^{q'}}$, and by (2) we get

$$\begin{aligned} \|e^{it\Delta}u_0\|_{L_{x,t}^q} &\leq \|u_0\|_{L_x^2(\mathbb{R}^n)} \sup_{\|f\|_{L_{x,t}^{q'}=1}} \left\| \int_{\mathbb{R}} e^{-it\Delta} f(\cdot, t) dt \right\|_{L_x^2(\mathbb{R}^n)} \\ &\lesssim \|u_0\|_{L_x^2(\mathbb{R}^n)} \sup_{\|f\|_{L_{x,t}^{q'}=1}} \left(\|f\|_{L_{x,t}^{q'}}^2 \right)^{1/2} = \|u_0\|_{L_x^2(\mathbb{R}^n)} \end{aligned}$$

which is our Strichartz estimate. $\square$

3.1. Comments on TT^* methods. The proof of (1) can be seen as an application of what is known as the TT^* method. This method, used in the study of dispersive PDE, revolves around the following result (see [1, Lemma 2.2]).

Lemma 3.3.1. *Let X be a Hilbert space and Y a Banach space. Then, if one of $T : X \rightarrow Y$, $T^* : Y^* \rightarrow X$, or $TT^* : Y^* \rightarrow Y$ is a bounded operator, then all three are.*

Put $T(u) = e^{it\Delta}u$ where $T : L_x^2(\mathbb{R}^n) \rightarrow L_{x,t}^q(\mathbb{R}^n \times \mathbb{R})$. Then, we see that $T^*(f) : L_{x,t}^{q'}(\mathbb{R}^n \times \mathbb{R}) \rightarrow L_x^2(\mathbb{R}^n)$ is given by

$$T^*(f) = \int_{\mathbb{R}} e^{-it\Delta} f(x, t) dt$$

and thus $TT^* : L_{x,t}^{q'}(\mathbb{R}^n \times \mathbb{R}) \rightarrow L_{x,t}^q(\mathbb{R}^n \times \mathbb{R})$ is given by

$$TT^*(g) = \int_{\mathbb{R}} e^{i(t-s)\Delta} f(x, s) ds.$$

In the preceding proof, we obtained the bound

$$\left\| \int_{\mathbb{R}} e^{-it\Delta} f(\cdot, t) dt \right\|_{L_x^2(\mathbb{R}^n)} \lesssim \|f\|_{L_{x,t}^{q'}},$$

showing that T^* is a bounded operator; by the TT^* method this implies bounds on the other two operators T and TT^* , with the boundedness of T being our Strichartz estimate (1).

More generally, TT^* methods come in various forms and are ubiquitous throughout harmonic analysis. For some history on their usage in different contexts of harmonic analysis, the reader is referred to [3, Section 1.1].

REFERENCES

- [1] Jean Ginibre and Giorgio Velo. Generalized Strichartz Inequalities for the Wave Equation. *J. Funct. Anal.*, 133(1):50–68, 1995.
- [2] Larry Guth. Real Analysis, Math 156. <https://math.mit.edu/~lguth/Math156.html>.
- [3] Bartosz Langowski, Mariusz Mirek, and Tomasz Z. Szarek. Discrete analogues in harmonic analysis: TT^* methods. arXiv:2606.07359, 2026.

- [4] Robert S. Strichartz. Restrictions of Fourier transforms to quadratic surfaces and decay of solutions of wave equations. *Duke Math. J.*, 44(3):705–714, 1977.